AMIRREZA VALAEI

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RESEARCH INTERESTS

- Reinforcement Learning
- Statistics
- Optimization

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) — Daejeon, South Korea

- M.Sc. in **Electrical Engineering**, U-AIM Lab — Sep 2026 – Jun 2028 *(expected)*
- *Advisor:* Prof. Chang D. Yoo

Sharif University of Technology — Tehran, Iran

- B.Sc. in **Electrical Engineering** — GPA: **17.65/20 (3.7/4.0)** — Sep 2021 – Jun 2026
- Minor in **Applied Mathematics** — Feb 2023 – Present
- **B.Sc. Thesis**: Numerical Optimization in TRPO — Prof. Hamed Shah-Mansouri

Allameh Jafari High School (NODET) — Marand, Iran

- **Mathematics and Physics** — GPA: **19.23/20** — 2018 – 2021

PUBLICATIONS & PREPRINTS

- **Controlling Gender Bias in Retrieval** — In Proceedings of ECIR 2026 — arXiv
  University of Amsterdam
  Authors: Amirabbas Afzali*, **Amirreza Valaei***, Iman Ahmadi, Mohammad Aliannejadi (*Equal contribution)
  Converted the Backpack LM into an encoder, enabling bias mitigation for ranking tasks.
- **Second-Order Policy Gradient Method for the LQR** — Engineering Applications of AI — arXiv
  Max Planck Institute for Software Systems
  Authors: **Amirreza Valaei**, Arash Bahari Kordabad, Sadegh Soudjani
  Designed an algorithm for exact second-order updates in deterministic policy-gradient methods for LQR.

OTHER RESEARCH EXPERIENCE

Sharif University of Technology

- **Numerical Optimization in Trust Region Policy Optimization** — Oct 2024 – Aug 2025
  Supervisor: Prof. Hamed Shah-Mansouri
  Analyzed conjugate-gradient updates in TRPO; reproduced baselines and ran controlled ablations.
- **Robust Reinforcement Learning** — Apr 2025 – Jan 2026
  Supervisor: Prof. Sajjad Amini
  Led a small research group on robust RL, conducted literature reviews, and implemented algorithms.

SELECTED ADVANCED COURSES

- **M.Sc., KAIST**: *Advanced Deep Learning*, *Advanced Deep Reinforcement Learning*, *Foundation Models*, *CS336: Language Modeling from Scratch (Stanford)*†
  † Audited `or` self-studied — *Italicized courses indicate current enrollment or study.*
- **B.Sc., Sharif University of Technology**:
  - *Undergraduate-level*: Linear Algebra – 19.8/20, Machine Learning – 19.5/20, Signals and Systems – 19.2/20, Convex Optimization – 17/20, Game Theory – 17.8/20, Numerical Optimization – 19/20, Mathematical Analysis I - 18/20, Differential Geometry - 18/20, Mathematical Analysis II - 18.5/20, Stochastic Processes - 19.2/20

– *Graduate-level*: Deep Learning – 17.2/20, Deep Reinforcement Learning – 20/20, Convex Optimization II – 18.7/20, High-Dimensional Probability Analysis – 19.3/20, Online Learning†, Information Theory – Statistics – and Learning – 19.5/20, Probabilistic Graphical Models - 19.3/20

SELECTED ADVANCED COURSE PROJECTS

- Poster presentation on "Robust Reinforcement Learning" and related projects [code and report]
- Implementation of the GANBERT model for a text classification task [code and report]
- High-dimensional analysis of the Neural Tangent Kernel [slides and report]
- Expectation-Maximization for a mixture of Gaussians [code and report]
- Lower-bound for non-stationary bandit algorithm [slides]
- Multiple deep learning course projects [code]

TEACHING EXPERIENCE

Teaching Assistant, Sharif University of Technology (*More information is available on my* teaching page)

- **Undergraduate Courses**: Probability and Statistics (Sp'23, Sp'24, Fa'24), Signals and Systems (Fa'23, Sp'24), Linear Algebra (Fa'23, Sp'24), Machine Learning (Fa'23, Sp'24, Fa'24 Lead TA, Sp'25, Fa'25), Convex Optimization I (Fa'24, Sp'25), Reinforcement Learning (Fa'25)
- **Graduate Courses**: Deep Learning (Fa'24), Deep Reinforcement Learning (Sp'25) High-Dimensional Probability Analysis (Fa'25) High-Dimensional Statistics (Fa'25)

VOLUNTEER EXPERIENCE

- **Scientific Associate Principal**, Student Association, Sharif University (Resana) Jun 2023 – Jul 2024
  Resana is the scientific community of the Electrical Engineering Department at Sharif University of Technology.
- **Organizer**, First AIVengers Summer School Sep 2023
  Planned and managed a global AI/ML student event featuring Ph.D. tutors from leading institutions.
- **Central Council Member**, ReACT 2024 Dec 2024
  Helped organize a large-scale event with 1,500+ attendees and internationally recognized researchers.
- **TEDx Sharif University of Technology** TEDxSUT Nov 2025
  Assisted in organizing and managing the TEDx event at Sharif University of Technology.

WORK EXPERIENCE

Tabdeal Aug 2024 – Jan 2025
Software Engineering Intern (On-site; Tehran, Iran)

- Assisted in backend Python development, optimized database search queries, and integrated internal information retrieval APIs for customer support tracking tools.

SKILLS AND INTERESTS

- **Programming:** Python (PyTorch, OpenCV, LangChain, Django), MATLAB, C++, Java, MySQL
- **Languages:** Persian, English, Turkish, Russian (Elementary), German (Beginner)
- **Interests:** Chess, Cinema, Soccer, Literature

HONORS & CERTIFICATIONS

- **Ranked 51st/165,000** in the Mathematics & Physics University Entrance Exam 2021
- Ranked **3rd** in the Control Systems cohort of Electrical Engineering at Sharif University 2026
- IELTS 7.5 (L:8, R:8, W:6.5, S:7) (verification link) 2025